SIC-Agents: Benchmarking and Building an Adaptive Simulator for Pediatric Serious Illness Communication Training

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Abstract

Pediatric serious illness communication (SIC) is critically important, yet scalable communication training for clinicians remains limited. Compared with other dialogue simulation settings, pediatric SIC poses additional challenges, including multi-party interactions, response to parental distress and strong dependence on feedback dynamics. Existing LLM-based simulators optimize generic dialogue quality rather than curriculum-contingent behavior required for effective SIC training. In collaboration with educators and pediatric clinicians, we introduce the first benchmark suite and simulation framework tailored to pediatric SIC training. Our benchmarks, PitfallBench and DialogueBench, evaluate simulators both at the turn-level and across full dialogues. We further propose SIC-Agents, a self-improving framework that generates a clinician-editable skill document to guide simulator behavior. Our experiments show that SIC-Agents outperforms static expert prompting. To support future research, we release our benchmarks[1] for parent simulation in pediatric SIC.

1 Introduction

When baby O was transferred to a new hospital, the medical team informed her family that brain imaging showed "evidence" of a terminal genetic disease. Medical facts were delivered, but the clinicians failed to acknowledge the emotional impact or meaning, leaving the family terrified, feeling pushed aside, and wondering whether their baby was going to die.[2] Experiences like this illustrate that serious illness communication[3] (SIC) requires more than the delivery of medical information alone. Yet, despite the critical importance of these communication skills, pediatric clinicians consistently report feeling underprepared (File et al., 2014; Wolfe et al., 2016; Rossfeld et al., 2018; Goloff et al., 2026; Dean et al., 2019).

Current SIC training relies largely on resource-intensive, difficult-to-scale standardized patient (SP) simulations, where an actor portrays a patient or parent and an observer provides feedback (Cleland et al., 2009; Back et al., 2007). While large language model (LLM) simulators offer a promising scalable alternative by allowing clinicians to engage in repeated, on-demand simulated conversations, and have been explored for psychotherapy, diagnostic interviewing, and adult palliative care communication (Tu et al., 2024; Wang et al., 2024, 2026; Haut et al., 2025; Sehgal et al., 2025), pediatric SIC differs significantly from these settings. In pediatrics, clinicians must navigate complex multi-party communication, respond to parental distress, communicate prognostic uncertainty, and support surrogate decision-making while maintaining trust during moments of vulnerability (van Breemen et al., 2020; Henderson and Boss, 2024).

[1]https://github.com/Beikewzh/sic-benchmarks

[2]Based on a real parent experience contributed by a parent-partner coauthor; full vignette in Appendix A.

[3]SIC refers to the communication skills required to care for patients and families facing serious or life-limiting illness (Feudtner, 2007; Bernacki and Block, 2014; Jacobsen et al., 2022; Henderson and Boss, 2024).

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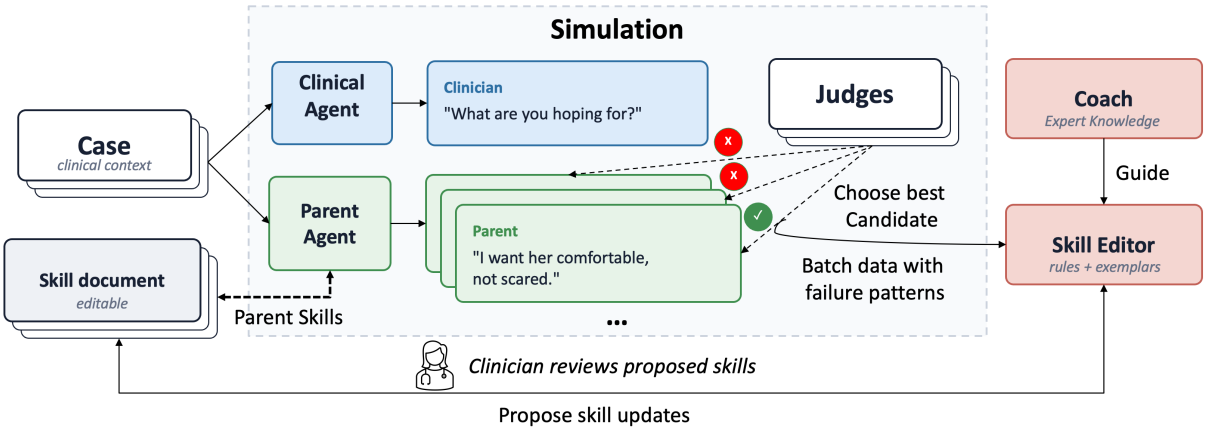


Figure 1: SIC-Agents self-improvement loop. A parent agent generates candidate responses conditioned on the case, dialogue history, and editable skill document. LLM judges select responses and provide feedback, which a Skill Editor uses to revise the skill document under curriculum guidance from a Coach agent. Clinicians and human instructors can audit and edit both the skill document and revisions.

This pediatric distinction exposes an important gap for the natural language processing (NLP) community. Existing medical dialogue simulators primarily emphasize information exchange, role adherence, or task completion. Pediatric SIC is a feedback-dependent interaction: clinician language continuously shapes parental emotion, trust, and engagement, which in turn alters subsequent conversational behavior (Cleland et al., 2009; Lewis et al., 2017). Crucially, this feedback-dependent interaction is important not only for accurately modeling clinical communication, but also for the underlying educational learning model itself; acquiring durable communication skills requires immediate feedback and iterative, deliberate practice (de Oliveira et al., 2024; Anders Ericsson, 2008). Effective simulation environments must therefore model not only clinical content, but also dynamically evolving interpersonal and emotional states across the course of a conversation. Despite their importance, such adaptive relational dynamics remain largely underexplored in NLP due to limited data availability, privacy constraints, and the clinician involvement required for development. More broadly, pediatric SIC introduces an important research area for adaptive dialogue systems operating in emotionally sensitive and multi-party settings.

To address this gap, we collaborated with educators and pediatric clinicians to develop benchmarks and simulation methods aligned with clinical training needs. We introduce two benchmarks and *SIC-Agents*, a self-improving parent simulator for pediatric SIC. We study whether a data-scarce simulator can improve through self-generated practice while remaining aligned with clinician-authored curriculum guidance. Unlike latent policy optimization approaches, SIC-Agents maintains a clinician-readable skill document that enables auditing, correction, and curriculum control. Because conversational quality depends not only on factual correctness but also on relational appropriateness, emotional responsiveness, and adaptation across conversational turns, evaluation is particularly challenging. We therefore propose benchmarks and clinician-defined evaluation dimensions to measure simulator behavior changes in response to supportive communication skills and common pitfalls. Together, the benchmarks and simulation framework position pediatric SIC as a testbed for emotionally grounded, curriculum-conditioned dialogue modeling in high-stakes settings.

Developed with educator and clinician co-authors, this paper makes three contributions:

1. **Evaluation Benchmarks and Metrics.** We introduce PITFALLBENCH and DIALOGUEBENCH, evaluation benchmarks for pediatric SIC, at the turn level and across full dialogues, respectively. The benchmarks compare supportive and suboptimal clinician communication strategies in matched scenarios. Through iterative expert refinement, we additionally define evaluation dimensions for emotional responsiveness, relational adaptation, and conversational grounding (§3).
2. **Inspectable Agentic Simulation.** We develop *SIC-Agents*, a multi-agent simulation framework that self-improves a clinician-editable skill document while using curricu-

lum guidance to reduce drift, i.e., deviations from medically and behaviorally appropriate parent responses (§4, §5) (Figure 1).

3. **Training Infrastructure for Deliberate Practice.** We integrate the simulator into a rapid-cycle deliberate-practice web-interface for SIC training, to support scalable evaluation and clinical education.

2 Background and Related Work

2.1 Identifying Learning Opportunities and Conversational Flow

Rather than strictly evaluating clinical competency, SP encounters are designed to track the flow of the conversation and identify critical learning opportunities where faculty can provide formative feedback (Cleland et al., 2009; Back et al., 2007; Paladino et al., 2019). In clinical education, learners are assessed on their ability to execute specific, clinically meaningful moves based on established serious illness frameworks (Ko et al., 2020). To operationalize these moves for targeted feedback, this work isolates the *trigger turn*: the exact point at which the learner commits one of these meaningful moves, acting as either a communication pitfall or its skilled alternative. Identifying these turns allows faculty to provide immediate feedback and, crucially, requires the simulated parent's response on the immediately following turn to dynamically adapt to what the clinician just said.

While established guides outline the core communication skills required for these conversations (Daubman et al., 2020; Breemen et al., 2024), effectively simulating them requires mapping where learners most commonly struggle (Meyer et al., 2009). Drawing on extensive faculty teaching experience within both pediatric and adult SIC programs, we empirically derived a taxonomy of ten common learner mistakes (pitfalls) distributed across two primary conversation phases: the ability to effectively share and support medical information, and the ability to explore goals and values and align the medical plan. For each pitfall, this expert-derived framework provides the core skill the learner forgets alongside the skilled-clinician alternative. Several of these observed pitfalls are uniquely tied to the pediatric relational dynamic, such as asking which treatments parents want rather than what they hope for, or framing what the clinical team will not do before framing what it will do (Breemen et al., 2024).

While the trigger turn provides a localized learning opportunity, the overarching conversational flow must also be maintained. A second key metric is the *across-session* judgment of whether the whole encounter stayed grounded and topically consistent over many turns. While relevant to human SP encounters, this metric is primarily used here to evaluate the difference between the output of an AI simulator compared to what a human actor would do, as an artificial dialogue can easily become disjointed even when individual turns are locally appropriate. A successful parent simulator must therefore be evaluated against both its contingent adaptation at the trigger turn and its ability to maintain a grounded, human-like dialogue across the entire encounter.

2.2 The Gap in Existing Conversational Agents

Communication-skill training is bottlenecked by scalable practice: durable acquisition requires repeated feedback-driven interaction that one-off SP workshops cannot provide (Anders Ericsson, 2008; Hunt et al., 2014; File et al., 2014). This has motivated LLM-based patient simulators built on earlier LLM-as-character work (Park et al., 2023; Shao et al., 2023). However, prior systems are evaluated mainly on generic realism metrics; none targets pediatric SIC, evaluates contingent adaptation at curriculum-defined pitfalls, or represents knowledge as a clinician-editable document.

Addressing this gap requires behavioral evaluation focused on specific capabilities rather than held-out accuracy (Ribeiro et al., 2020). Similar ideas have recently appeared in clinical evaluation through MATRIX (Lim et al., 2025). We apply this framework in pediatric SIC, where privacy constraints preclude large public corpora (Singhal et al., 2023; Luo et al., 2022). Accordingly, the benchmarks in §3 evaluate contingent adaptation at trigger turns and grounding across extended dialogue. A comparison with existing clinician-anchored dialogue benchmarks is in Table 1. Prior self-improving LLM frameworks such as Reflexion (Shinn et al., 2023), SELF-REFINE (Madaan et al., 2023), ACE (Zhang et al., 2025), and Constitutional AI (Bai et al., 2022) show iterative behavioral refinement without weight updates. However, generic self-improvement targets can improve surface dialogue quality while missing the curriculum-contingent behavior that makes a pediatric SIC simulator clinically valid. SIC-Agents therefore uses

these standard self-improvement components, but also embeds pediatric SIC curriculum feedback in a clinician-editable form (§4).

3 Datasets

Public pediatric serious-illness-conversation (SIC) dialogue data are effectively unavailable because of clinical confidentiality constraints (Singhal et al., 2023; Luo et al., 2022). Consequently, prior work has lacked both (i) realistic substrates for training pediatric-parent simulators and (ii) clinician-anchored evaluation resources for measuring simulator quality. To address this gap, clinician co-authors with extensive experience in SIC training identified two key evaluation targets (§2): *turn-level contingent adaptation* and *dialogue-level grounding*. Accordingly, we release two corresponding resources:

- **PITFALLBENCH**: a 1,000-item turn-level benchmark that evaluates whether a parent simulator reacts appropriately to clinician communication quality at a *trigger turn*; and
- **DIALOGUEBENCH**: a 152-dialogue gold benchmark that evaluates whether a simulator maintains coherent grounding and role consistency across a full multi-turn encounter. Accompanied by SIC-SimCorpus, an 8,391-dialogue silver corpus.

Both resources are constructed in close collaboration with pediatric SIC clinicians with training experience and are anchored in the Serious Illness Conversation Guide for Pediatrics curriculum (Daubman et al., 2020; Breemen et al., 2024). Unlike benchmarks based primarily on post hoc per-item annotation (Table 1), our resources define validity through an explicit, auditable acceptance criterion jointly developed and refined by clinicians and an independent LLM reviewer.

Benchmark	Domain	Gold	Expert
Patient-Ψ (Wang et al., 2024)	Psychotherapy	106	designed
PatientSim (Kyung et al., 2025)	General	37	checked
AMIE (Tu et al., 2024)	Diagnostic	149	rated
PitfallBench (ours)	Pediatric SIC	**1,000**	curriculum
DialogueBench (ours)	Pediatric SIC	**152**	curriculum

Table 1: Comparison with existing clinician-anchored dialogue benchmarks. *Gold* denotes the number of released evaluation items. Our benchmarks are grounded in an external pediatric SIC curriculum and refined under expert oversight.



Figure 2: PitfallBench uses a paired within-case design. The same case and dialogue history are instantiated twice at the trigger turn: once with a curriculum-defined pitfall move and once with its skilled alternative. The simulator must generate differentially appropriate parent responses under otherwise identical conversational conditions.

3.1 PITFALLBENCH

PitfallBench evaluates contingent adaptation at the level of a single *trigger turn*. The benchmark is motivated by a central pedagogical objective in pediatric SIC training: a clinician utterance that follows best practice should elicit a meaningfully different parent response than a clinically suboptimal utterance under the same conversational context.

Task formulation Each PITFALLBENCH item consists of a pediatric clinical case, a partial dialogue history up to a *trigger turn*, a curriculum-defined *pitfall* clinician move, its corresponding *skilled* alternative, and the expected parent reaction associated with each move (Figure 2). At evaluation time, the simulator receives the case context, the dialogue history, and one clinician move, and generates the next parent utterance. A judge LLM then classifies the generated response as `challenge_signal`, `reward_signal`, or `neither` relative to the curriculum-defined expected reaction. The benchmark therefore operationalizes contingent adaptation as a constrained three-way classification problem over simulator behavior.

Curriculum structure PitfallBench is organized around ten common learner pitfalls (Table 2) compiled by our pediatric SIC clinician co-authors (Appendix E). Each pitfall is paired with a *curriculum rule*, an expert-authored template that fixes the dialogue state, the suboptimal clinician move, its

skilled counterpart, and the *expected reaction label* that becomes the gold class (`challenge_signal` for the pitfall, `reward_signal` for the skilled alternative; per-pitfall sketches in Appendix Table 4). The cue–response contingencies these rules encode are documented in expert-annotated recordings of real pediatric care conferences (Olszewski et al., 2023), whose observed family responses correspond to the reward and challenge dynamics our items formalize.

Paired evaluation design The benchmark uses a paired within-case construction in which both the pitfall move and the skilled alternative are instantiated under the same dialogue state. This design controls for case-level variation and isolates the behavioral effect of the clinician move itself. It also reduces degenerate evaluation strategies: simulators that default to emotional escalation fail the skilled condition, while simulators that produce uniformly neutral responses fail the pitfall condition. Successful performance therefore requires contingent behavioral discrimination between the paired stimuli.

Gold-label construction A central methodological contribution of PITFALLBENCH is its criterion-based validation protocol. Rather than relying exclusively on per-item expert annotation, we define validity through an explicit acceptance criterion iteratively calibrated between pediatric SIC clinicians and an independent LLM reviewer from a different model family than the generator.

Candidate items are first generated from curriculum templates, after which the reviewer applies the current written acceptance criterion. Clinicians (Appendix E) then inspect reviewer decisions on a stratified sample. Disagreements trigger revisions to the criterion and generation prompts rather than manual editing of individual items, and failed items are regenerated and re-screened. Iteration terminates once reviewer–clinician agreement reaches 95% on a stratified validation sample of 50 items (5 per pitfall). The resulting frozen criterion is then applied uniformly across all released items, with every screening decision logged together with the criterion version. This protocol follows the screen-and-regenerate paradigm used in recent synthetic clinical benchmark construction (Wang et al., 2024; Kyung et al., 2025) while making the acceptance rule itself an explicit and auditable object.

Dataset composition The benchmark contains 100 underlying pediatric cases stratified across 12 pediatric care settings, child ages 0–18, multiple disease categories, and diverse family configurations. Each of the 1,000 items embeds a partial dialogue history of up to ten turns (median eight) leading into the trigger turn. An embedding-based diversity audit (Appendix C) shows broad semantic coverage across clinical scenarios while preserving tight within-case clustering for paired comparisons.

3.2 DIALOGUEBENCH

While PITFALLBENCH evaluates local contingent adaptation, DIALOGUEBENCH evaluates dialogue-level grounding across a full multi-turn pediatric SIC encounter.

Substrate generation Because no public pediatric SIC corpus exists, we first construct a synthetic dialogue substrate (SIC-SimCorpus) using a factorized, persona-conditioned generation pipeline. The pipeline varies eight LLM generator families, six pediatric disease categories, three trainee proficiency levels, and dialogue lengths ranging from 12–28 turns. Using multiple generator families reduces dependence on any single model distribution and improves stylistic diversity. Full generation statistics are in Appendix C.

ID	Pitfall	Core SIC skill
Phase 1: share and support information		
P1	Missed early emotion cue	Ask for permission
P2	Forgets to assess understanding	Assess understanding
P3	Information too complex	Headline framing
P4	Missed or combined empathy cues	Respond to emotion
P5	Stuck in validation	Move on after empathy
Phase 2: explore goals and align plan		
P6	Straight to code status	Elicit values first
P7	Single or too-broad value	Multiple specific values
P8	Cannot consolidate or align	Consolidate + recommend
P9	Asks treatments not values	Hopes / worries framing
P10	"Won't do" before "will do"	Lead with will

Table 2: The ten learner pitfalls covered by PitfallBench, grouped by curriculum phase, with the core SIC skill each pitfall compromises.

Gold/silver release structure Following the silver-standard methodology established in biomedical NLP (Rebholz-Schuhmann et al., 2010), we adopt a gold/silver release structure. The Silver tier consists of the full 8,391-dialogue synthetic corpus,

Case: 7 yo with progressive neurodegenerative disease; outpatient palliative-care consultation; mother present.
Clinician: I want to make sure we are on the same page about what the team is most worried about today.
Parent: I just keep hoping the new medication will start to work.
Clinician: It is hard to hold both hope and worry at the same time. Can you tell me what you have been most worried about lately?
Parent: I am worried that she is not going to be here much longer. (*long pause*) I do not know how to tell her brother.
[. . .]

Figure 3: Excerpt from a DIALOGUEBENCH. During evaluation, clinician turns are fixed and the simulator generates the parent turns. Responses are scored on Conversational Grounding and Topic Consistency.

released as a finetuning substrate, while the Gold tier consists of the final 152-dialogue evaluation benchmark reviewed by human experts. Gold-tier selection proceeds in two stages. An automated scoring pass first removes dialogues that fail a fixed threshold on a two-axis rubric measuring Conversational Grounding (responsiveness to the clinician's latest turn) and Topic Consistency (maintenance of a coherent concern). The rubrics are defined by pediatric clinicians. Pediatric SIC clinicians then review the surviving dialogues for clinical realism using the same frozen acceptance criterion applied in PITFALLBENCH. Using a shared criterion ensures that both benchmarks inherit the same operational definition of clinical validity.

Evaluation protocol During evaluation, the clinician turns from a benchmark dialogue are held fixed as a script. The simulator generates parent responses in place of the original turns, and generated responses are scored by two independent LLM judges on a two-axis 1–3 rubric for *Conversational Grounding* and *Topic Consistency*. The judges are calibrated against annotations from 11 pediatric clinicians; see Appendix D. Because the clinician script is identical across systems, evaluation is performed under matched conversational conditions, enabling direct item-level comparison between simulators. The original parent turns additionally serve as a reference condition in the matched comparisons reported in §5.

4 SIC-Agents

SIC-Agents is a curriculum-conditioned self-improving parent simulation framework (Figure 1) designed with two objectives: (1) to generate interactional training evidence without requiring access to private clinical dialogue data, and (2) to preserve interpretability by representing adaptive parent behavior in an editable natural-language skill document rather than in latent model parameters.

The encounter is anchored by a fixed clinician agent guided by the same pediatric SIC curriculum (Daubman et al., 2020; Breemen et al., 2024). Its role is to provide a stable learner-side interlocutor that follows the SIC guide well enough to create coherent practice turns and, in self-improvement runs, to surface the kinds of clinician moves the parent simulator must react to.

The parent agent is conditioned on the pediatric case, dialogue history, and a mutable skill document S. The skill document serves as an editable behavioral specification, encoding six expression patterns (Appendix F), exemplar utterances, transition guidance, and constraints on length, role fidelity, emotional calibration, and off-topic generation. Because S is represented in natural language, clinical experts can inspect and revise it directly, and the parent agent can use the revised document in subsequent simulation cycles.

In each simulation cycle, the parent agent samples K candidate responses at every parent turn t ($K{=}3$ in our experiments; Appendix H). Invalid generations are removed via a hard filter, after which the remaining candidates are evaluated by an LLM-judge ensemble along the dimensions of *Conversational Grounding* and *Topic Consistency* (Appendix D). The highest-scoring candidate is selected for inclusion in the dialogue, while both selected and rejected candidates are retained as contrastive evidence for the Skill Editor. This process produces a lightweight source of self-generated preference data that exposes both the strengths and failure modes of the current skill document S.

Conditioned on this batch evidence, the Skill Editor proposes localized modifications to S, such as adding exemplars, refining constraints, revising response patterns, or introducing previously missing interaction strategies. Consequently, adaptation occurs over the interpretable skill document rather than the underlying LLM parameters. Across simulation cycles, S incrementally accumulates interaction-derived behavioral rules and exemplars from simulated practice.

During self-improvement, the Skill Editor receives two complementary feedback streams: Judge-derived dialogue feedback and Coach-derived curriculum feedback. The judge ensemble

identifies interaction-level deficiencies, including failures in conversational grounding, topic maintenance, response length, and role fidelity. However, these scores do not capture the pedagogical context of the interaction. The Coach provides this missing curriculum-level supervision by specifying which learner pitfall a clinician behavior instantiates, which SIC skill is being exercised, and what parent reaction should appropriately follow. This information is derived from the same pediatric SIC curriculum and includes the learner pitfall, target SIC skill, and expected parent response trajectory. During evaluation, neither feedback stream is available to the parent agent, which operates solely from the frozen skill document S.

The resulting skill document powers an open-source training platform where learners engage the parent agent in multi-turn interactions with curriculum-grounded feedback. Faculty can route challenging cases back through the Skill Editor, enabling iterative refinement from deployment data. More broadly, SIC-Agents supports confidential, data-scarce domains by improving an interpretable policy via curriculum-guided self-improvement, and extends naturally to other feedback-intensive settings such as mental-health counseling, crisis intervention, and social-work training.

5 Experiments

We conduct experiments on SIC-Agents as a feedback-distillation framework operating over a frozen skill document S. At evaluation time, the parent agent receives no external feedback and generates responses solely conditioned on S. We investigate whether static prompting is sufficient, whether self-improvement without curriculum supervision is adequate, and whether curriculum-level supervision is necessary to preserve clinically contingent behavior.

We evaluate on PITFALLBENCH (§3.1), which measures whether the parent agent reacts appropriately to a curriculum-defined pitfall or skilled clinician move at a matched trigger turn, and on Gold DIALOGUEBENCH (§3.2), which measures dialogue-level grounding and topic consistency. Our primary outcome is PITFALLBENCH accuracy, because it tests the curriculum-contingent behavior that generic dialogue feedback does not label. We also report DIALOGUEBENCH Grounding and a 0–1 composite that averages PitfallBench accuracy with grounding rescaled to 0–1 as secondary summaries. Because the generation loop itself uses grounding/topic judges, DialogueBench is partly aligned with the generic feedback stream. We therefore interpret it as a check that dialogue-quality feedback was retained in the skill document, not as an independent proof of clinical alignment. Table 3 gives the headline results; full per-condition breakdowns are in Appendix I.

Experimental settings. We compare five deployable conditions that isolate where the curriculum enters the system:

- *No loop*: a deliberately minimal parent seed; an expert prompt that places expert-written SIC principles directly in the parent's context; and a full-taxonomy prompt that further adds the explicit P1–P10 taxonomy and expected reactions to the expert prompt (Appendix H).
- *Self-improvement*: two systems begin from the same minimal seed and run five edit cycles. The *no-curriculum* ablation keeps candidate sampling, a panel of judges, and the Skill Editor, but removes the Coach signal. The full system adds the Coach signal.

Curriculum-signal isolation. Because the curriculum signal derives from the same pitfall taxonomy used to construct PitfallBench, we restrict it to self-improvement. During training, the clinician agent uses the same curriculum-guided configuration across conditions, so the ablation isolates the Coach and Skill Editor rather than the simulated encounter itself (Appendix H). During benchmark evaluation, clinician turns are fixed scripts from PitfallBench or DialogueBench, and all deployable parent agents run from the frozen skill document only; the parent never sees the item label or expected reaction. An oracle variant that reveals the signal at inference is reported only as an upper bound in Appendix I.

5.1 Iterative self-improvement outperforms static expert prompting

The full self-improvement condition is the strongest on the primary PITFALLBENCH outcome and improves DIALOGUEBENCH Grounding, yielding the highest composite score (Table 3). Over three independent seeds, the full system attains PitfallBench 0.78 ± 0.03 and Grounding 2.96 ± 0.02. The expert-prompt baseline is the key no-loop control. It contains an expert-written, principle-level summary of how a parent should react to supportive versus poorly timed clinician

Condition	Primary	Secondary	
	PitfallB. ↑	DialogueB. ↑	Comp. ↑
No loop			
Minimal seed	0.66	2.52	0.71
Expert prompt	0.73	2.28	0.69
Full-taxonomy	0.75	2.32	0.71
Self-improvement			
No curriculum	0.52	2.79	0.71
SIC-Agents	**0.78**	**2.96**	**0.88**

Table 3: Main results on PitfallBench and DialogueBench. All self-improvement conditions share the same minimal seed. PitfallBench accuracy and the composite are reported on a 0–1 scale; DialogueBench reports Grounding on the original 1–3 rubric. The composite averages PitfallBench accuracy with Grounding rescaled to 0–1.

communication, but it does not contain the P1–P10 taxonomy or per-item expected reactions (Appendix H). This improves pitfall resistance (0.85 versus 0.75 for the seed; Appendix I) but produces the weakest grounding score of any condition. The full-taxonomy prompt adds the explicit P1–P10 taxonomy and expected reactions on top of the expert prompt, with no self-improvement loop. It changes little from the expert prompt, which already carries these ideas implicitly, and remains below the full system on all three metrics, most clearly on grounding (2.32 versus 2.96). Handing the parent the benchmark's taxonomy is therefore not sufficient: the evolved skill document is not simply restating the benchmark's expected behaviors. The full system does not win by making the parent prompt longer or more expert-sounding; it wins by changing how practice evidence and curriculum feedback are converted into an editable skill document.

This distinction matters for a clinically deployed simulator. A prompt written once can encode only the contingencies its authors anticipated, and adding more expert-authored prose can make the parent sound as if it is reciting clinical concepts rather than inhabiting a parent role. The self-improvement loop instead generates its own training evidence: it samples multiple parent candidates, keeps the successful and unsuccessful responses as contrastive examples, and asks the Skill Editor to turn the observed failure pattern into a small text edit. We find that a data-scarce clinical simulator can improve by generating practice cases and examples for an editable policy, rather than relying on a fixed hand-crafted prompt.

5.2 Curriculum guidance prevents drift

The curriculum ablation tests whether the judge ensemble feedback is sufficient. Both self-improvement conditions start from the same seed, generate the same number of candidates, use the same judges, and run the same five edit cycles. The no-curriculum condition is not a system-without-feedback baseline; the judge ensemble remains. The only removed information is the Coach's pitfall taxonomy and expected parent reactions.

Without the Coach, the loop succeeds on the feedback it receives. DialogueBench Grounding remains high and improves across cycles (Figure 4), which is expected because grounding is part of the judge-derived training signal. The failure appears when the same system is evaluated on curriculum-anchored contingent behavior: PitfallBench accuracy drops below the initial seed, with the largest loss in skilled-reward responses. The result is therefore not that DialogueBench performance alone proves improvement; it shows the opposite limitation. Generic dialogue feedback can be distilled into a fluent simulator while clinically decisive trigger-turn behavior erodes.

The full system reduces this drift because the Coach adds a clinical interpretation to the failures the editor already sees. A judge can say that a response is grounded; the Coach can say that the clinician just committed a learner pitfall, that the missing SIC skill is permission or values elicitation, and that the parent should push back, grant permission, stay uncertain, or reveal a value. The Skill Editor writes that target into the skill document as a rule, constraint, or exemplar. The damage from removing this signal concentrates in pediatric-specific pitfalls P8 and P10, where the no-curriculum system almost never rewards the skilled alternative (Appendix Table 6).

Two caveats bound the interpretation. First, pitfall-resistance remains easier than skilled-reward even for the full system: the skill document learns to push back on problematic clinician moves more readily than to reward skilled ones. Second, the trajectory analysis uses a smaller evaluation subset than the main table, so we interpret the across-cycle pattern rather than cycle-to-cycle changes.

Judge robustness. Because the parent agent, Skill Editor, and paper judges are all Claude models, we re-scored the key conditions on PITFALLBENCH with three non-Claude judges (GPT-5, DeepSeek R1, and Qwen3). The ordering full sys-

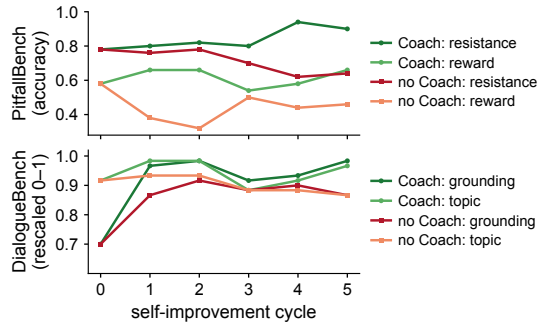


Figure 4: Self-improvement trajectories from the same minimal seed. **Top:** PitfallBench sub-metrics. Curriculum guidance preserves or improves pitfall-resistance and skilled-reward; without it, both decline. **Bottom:** DialogueBench Grounding and Topic Consistency (rescaled to 0–1) remain high in both conditions. The divergence between the two panels is the empirical signature of clinical drift.

tem > expert prompt > no-curriculum holds under all three judges with small score differences (Appendix I, Table 7), so the gains are not an artifact of Claude judging Claude.

6 Conclusion

We introduced PITFALLBENCH and DIALOGUEBENCH, two clinician-anchored pediatric SIC benchmarks, together with SIC-Agents, a self-improving framework for parent simulation. Our central finding is that generic dialogue optimization alone is insufficient for clinically grounded simulation: without curriculum guidance, self-improvement improves surface dialogue quality while degrading clinically critical contingent behaviors. The important design choice is therefore not a more complex optimizer, but a loop that can generate practice evidence, use judge contrasts to identify editable failures, use expert curriculum labels to interpret clinically meaningful failures, and write the result back into a skill document clinicians can inspect. SIC-Agents suggests a practical path for high-stakes, data-scarce domains where supervised fine-tuning is infeasible but hand-crafted prompts are too brittle. Future work can extend this framework with fully human-in-the-loop edit approval and distillation of the skill document into smaller models for resource-constrained clinical settings.

Limitations

Clinician validation is criterion-level, not per-item. Expert validation of our benchmarks is based on stratified samples and criterion-level review rather than individual annotation of all 1,000 PitfallBench items or all 152 DialogueBench dialogues. Expert effort was concentrated on defining and iteratively refining a high-fidelity acceptance criterion that governs dataset inclusion (§3.1), through repeated expert–reviewer adjudication and screen-and-regenerate cycles until convergence, yielding a stable, expert-anchored, and auditable admission rule. Extending expert validation to per-item annotation is a natural direction for future work.

Scope of evaluation. We present controlled simulation-based evaluations on two benchmarks designed to isolate communication behaviors in pediatric SIC. Our experiments do not evaluate trainee learning outcomes, skill transfer, or downstream clinical impact; these prospective questions require IRB-governed trainee studies, which we leave to future work.

Synthetic data and simulator fidelity. Due to the confidentiality of pediatric SIC conversations, the DialogueBench substrate is synthetically generated. Multi-vendor generation and an expert realism gate reduce single-model artifacts, and the cue–response contingencies our items encode are documented in expert-annotated recordings of real pediatric care conferences (Olszewski et al., 2023). Nonetheless, the synthetic dialogues have not been directly validated against the distribution, emotional dynamics, or long-tail complexity of real pediatric SIC conversations: the benchmarks evaluate performance against a curated curriculum rather than real transcripts. Their fidelity target is functional, providing enough realism to elicit and reinforce the behaviors the curriculum teaches (Hamstra et al., 2014); these are limitations of scope and external validity rather than of the framework itself.

Cultural and linguistic generalization. The curriculum, pitfall taxonomy, and communication patterns are grounded in a single English-language pediatric SIC curriculum and validated by one expert panel. The curriculum deliberately teaches culture-general elicitation skills rather than culture-specific scripts (Rosenberg et al., 2017), and the cases vary family background and configuration, but whether these behaviors transfer across languages, cultures, and healthcare systems has not been tested and re-

mains an open and important direction for future work.

Single model family for the simulator. The parent agent and Skill Editor are built exclusively on Claude models. Although re-scoring with non-Claude judges preserves the condition ordering (Appendix I, Table 7), the simulator's behavior itself may carry model-family-specific stylistic and distributional biases, and replicating the self-improvement loop across model families is future work.

Clinical-decision-making boundary. SIC-Agents is intended strictly as a communication training simulator for pediatric SIC contexts. It does not provide clinical decision support, does not offer medical advice, and must not be used to inform real-world patient care.

Ethical Considerations

This work develops LLM-based simulators for pediatric serious illness communication (SIC), a high-stakes and emotionally sensitive clinical setting. The system is intended for clinician training and research, not for medical decision-making or direct patient interaction.

The project has been reviewed and approved by our institution's research ethics board (REB). This paper uses no patient data and involves no patients or trainees: all cases are synthetic and de-identified, and clinicians rated synthetic dialogues only. The approval also covers the subsequent trainee study in the hospital.

The benchmarks, evaluation criteria, and simulator design were developed in close collaboration with pediatric clinicians, including clinician co-authors. SIC-Agents is designed to prioritize curriculum-consistent and clinically grounded behavior through curriculum guidance, constitutional filtering, and inspectable clinician-readable policies. However, LLMs may still generate unsafe, biased, or emotionally inappropriate responses and may inherit demographic and cultural biases from training data, so simulator behavior may not generalize equally across populations and human oversight remains necessary.

Simulated parents cannot fully capture the diversity of real families facing serious illness. The simulator complements rather than replaces supervised clinical training, and no system output is medical advice or a basis for patient care.

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A Parent Partner Vignette

The following account is shared by a parent partner and motivates the opening of §1. Identifying details have been generalized.

When baby O became sick, she was lethargic, completely limp. Her mother A knew in her gut that something was gravely wrong. They lived in a remote community. The staff at the local clinic refused to do further testing, concluding "I have no worries, keep monitoring her temperature at home." On the third day of being repeatedly refused further testing, A made the decision to seek a second opinion, an hour drive from the community. The ER doctor listened to O's lungs and concluded "I would treat as pneumonia." A few days later O was not getting better, even on antibiotics. O began to show signs of seizures, her eyes starting to roll back in her head. An ER in a bigger town, a three-hour drive from the community, admitted O. After further X-rays and ultrasounds came back normal, the staff grew concerned that O might be affected by a rare disease and quickly arranged for her to be flown to the children's hospital in the urban centre. The transfer note indicated that the baby was potentially at risk for a terminal genetic illness known as Cree Leukoencephalopathy (CLE), although A and her husband J were not aware of what that really meant.

At the children's hospital, O was admitted under the care of a medical team. Following an MRI, A remembers a resident coming by to tell them that they had the "results" from the scan but offering no further explanation, leaving them in the dark awaiting news from the attending physician. A and J's stress mounted from knowing that there were "results" but not being told what they were. When the attending physician eventually arrived, they were informed that the MRI showed "evidence" of CLE, but that the diagnosis could not be confirmed without specialized blood tests. The team was handing off the case to another specialist, as this was their domain of expertise, and the family would receive a phone call in several weeks. A and J were left wondering "does she have it?", "could it be something else?", "what if they're wrong?", "what does it mean to 'see evidence' of a terminal illness?", and many other questions. The exchange with the teams had been brief and made A and J feel pushed aside. The doctors did not invite their questions or ask them what they needed to know. A would have liked to hear "Is there something you don't understand?" or "What are you thinking?" Instead, A and J were left scared and unprepared for what was going to happen next. Was their baby going to die?

B Benchmark Examples

To make the unit of each benchmark concrete, we show one released item of each kind: a PitfallBench item (a single trigger turn) and a DialogueBench dialogue (a full encounter).

B.1 A PitfallBench item

Each item instantiates one shared case and history twice at the trigger turn, once with the curriculum-named pitfall move and once with its skilled alternative; a faithful parent simulator must react in opposite directions. Item `BENCH_PHASE2_001` below covers pitfall P6 ("straight to code status").

Case (neurology; school-age; two parents): Lily Chen, a previously well 7-year-old in rapid neurological decline; MRI shows progressive leukoencephalopathy, genetic testing pending. **Family:** "We just want to know what's happening to our little girl. There has to be something you can do."

History (up to the trigger turn):
Clinician: Would it be okay if we talk about what we have been finding?
Parent: Yes, please. We just need to know what's going on.
Clinician: Before I share, what is your understanding of what has been happening?
Parent: They said white matter on the MRI, and we keep being told to wait. She was reading chapter books six months ago.

Pitfall move (P6): "If Lily's heart were to stop, would you want us to do CPR, a breathing tube, machines to bring her back?" → expected `CHALLENGE` (escalates, doubles down on intervention).

Skilled move: "Before we talk about what comes next, what are you most hoping for her, and what worries you most?" → expected `REWARD` (becomes collaborative, shares a hope or value).

Provenance: pitfall `P6_straight_to_code_status`; core skill *eliciting values*; anchored to the curriculum's Phase-2 pitfalls; case drawn from the gold corpus.

B.2 A DialogueBench dialogue

At evaluation the clinician turns are a fixed script and the simulator produces the parent turns, each scored on Grounding and Topic Consistency (1–3). The excerpt below annotates each parent turn with its (Grounding, Topic) scores.

ID	Curriculum rule (trigger → pitfall vs. skilled)
Phase 1: share and support information	
P1	Parent gives an early emotional cue; pitfall proceeds to content, skilled asks permission and names the cue.
P2	New information just delivered; pitfall moves on, skilled asks the parent to restate it.
P3	Complex information to convey; pitfall front-loads detail, skilled leads with a headline.
P4	Parent shows empathy cues; pitfall misses or collapses them, skilled responds to each emotion cue distinctly.
P5	Parent just validated; pitfall keeps re-validating, skilled briefly acknowledges and shifts forward.
Phase 2: explore goals and align plan	
P6	Goals-of-care territory; pitfall jumps to code status, skilled first elicits hopes and worries.
P7	Exploring what matters; pitfall accepts one or a broad value, skilled surfaces several concrete values.
P8	Values elicited; pitfall leaves them unintegrated, skilled consolidates and recommends.
P9	Moving toward the plan; pitfall asks the parent to choose treatments, skilled uses hopes/worries framing.
P10	Sharing the plan; pitfall opens with what will *not* be done, skilled leads with what *will* be done.

Table 4: Per-pitfall curriculum rules for PitfallBench, authored by our clinician co-authors (Appendix E). Expected parent reaction is `challenge_signal` for the pitfall move and `reward_signal` for the skilled move in every row; the pitfall name and core SIC skill are in main-text Table 2.

Case: school-age child (Aidan) with sickle cell disease after a stroke; mother (Maeve).

Clinician: Maeve, I want to make sure you understand everything. How are you holding up?
Parent: [looks down] I'm trying. It's just a lot. (G 3, T 3)
Clinician: Aidan's sickle cell disease is something he was born with, a genetic trait that affects the shape of his red blood cells.
Parent: [quietly] But we don't have this in our family. (G 2, T 3)
Clinician: It can appear even without a known family history, if both parents carry the trait. Genetics can give some clarity.
Parent: [nods] Okay. (G 3, T 3)
[. . .]

Over six parent turns: mean Grounding 2.5, mean Topic 3.0, which clears the Gold gate.

C Benchmark Diversity

PitfallBench. Diversity in PitfallBench lives at the case level. The bench draws on 100 distinct pediatric cases, each instantiated against all ten pitfalls to give 1,000 items (100 per pitfall); the cases are stratified across twelve pediatric care settings (neonatal, cardiac, and general intensive care; oncology; neuro-oncology; bone-marrow transplant; neurology; palliative care; pulmonary; renal; gastroenterology; and emergency), child age 0–18, and a range of family configurations. Figure 5 projects the 1,000 items with UMAP over Gemini text embeddings, and they spread broadly across the space rather than collapsing onto a handful of templates. Within a case the ten paired moves share its clinical content by design, so the within-case comparison isolates the clinician move (§3.1); the breadth comes from the 100 distinct cases.

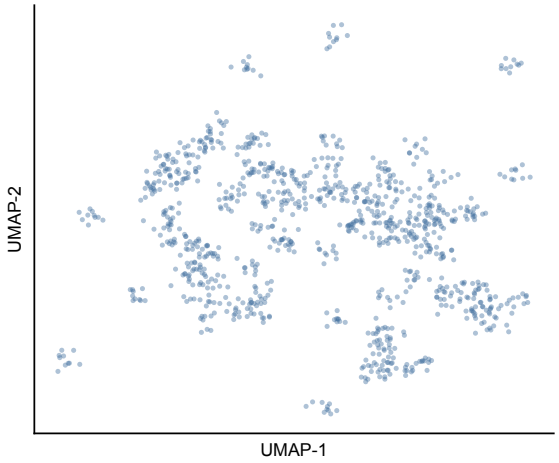


Figure 5: PitfallBench diversity. UMAP of the 1,000 items in Gemini embedding space; the items spread broadly across the space, covering the 100 cases and their twelve care settings.

DialogueBench. DialogueBench inherits its diversity from how the corpus is built. The 8,391 released dialogues come from a factorized, persona-conditioned pipeline that crosses eight LLM generators (GPT-4o, GPT-5, GPT-5.1, Claude Sonnet 4, Gemini 3 Pro, Llama-3.3-70B, Qwen3-235B, and Kimi-K2, each contributing 723–1,117 dialogues), three near-evenly split trainee proficiency levels (beginner, proficient, expert), and dialogue length (12–28 turns, mean 20); the underlying cases span six pediatric disease categories. Drawing each dialogue from a different model keeps the corpus from being a single-model artifact. Figure 6 projects the parent-response embeddings of the 8,391-dialogue corpus with UMAP; they fill the space broadly rather than collapsing into a few modes. The Silver tier releases the full corpus; the Gold set is the 152 dialogues that pass the rubric screen and the clinical-realism gate (§3.2).

D DialogueBench Rubric and Judge Calibration

Rubric. DialogueBench scores each parent turn on two axes, each on a 1–3 scale. *Conversational*

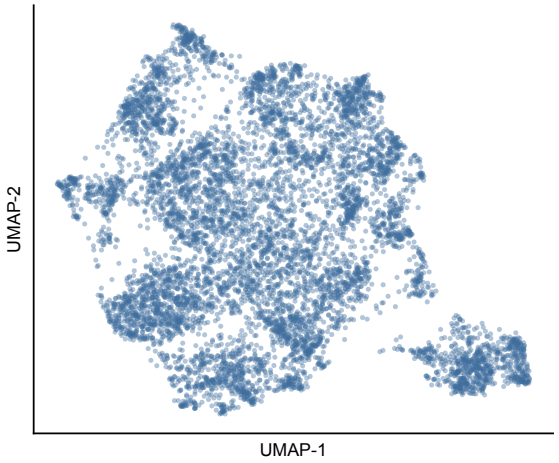


Figure 6: DialogueBench diversity. UMAP of the parent-response embeddings from the 8,391-dialogue corpus (text-embedding space, cosine metric). The responses spread broadly across the space.

Grounding measures whether the reply connects to what the clinician just said: 1, scripted (ignores the clinician's words); 2, relevant but generic; 3, directly references the clinician's last turn. *Topic Consistency* measures focus within the reply: 1, scattered across unrelated topics; 2, two connected concerns; 3, a single focused worry. A turn also hard-fails if it introduces facts or jargon the clinician never raised, runs over a brief length, or ignores a direct question. The two axes were settled with our clinical collaborators; a candidate emotional-authenticity axis was dropped because its expert ratings could not be reliably differentiated (§3.2).

Judge calibration. The DialogueBench rubric judge is supporting instrumentation, not the basis of the bench's validity (§3.2); we record its agreement with expert ratings here for completeness. The judge is calibrated against 119 expert ratings collected from 11 pediatric clinicians on a stratified sample of 28 DialogueBench-source dialogues, scored with the same two-axis rubric used at evaluation time. Per-axis Spearman correlation between the judge and the pooled expert ratings is $\rho_{\text{Grounding}}=0.71$ and $\rho_{\text{Topic Consistency}}=0.60$, in line with judge–human agreement reported in adjacent patient-simulator work (Wang et al., 2024; Kyung et al., 2025). These ratings were collected on full dialogues with the dialogue-level rubric and are independent of the PitfallBench classification labels, which are anchored on the curriculum provenance fields (§3.1) rather than on this judge.

E Expert Panel and Demographics

Pediatric SIC sits at the intersection of medical, psychosocial, and educational expertise, so the resources in this paper were built and validated by a multidisciplinary expert panel rather than by physicians alone. The panel comprises four pediatric physicians whose practice centres on serious-illness and palliative communication, one clinical social worker, and one clinician-educator with a background in communication-skills teaching. All members practise and teach in English-language pediatric settings, which is also the intended deployment context of the training interface; extending the resources to other languages and health systems is left to future work (§6).

The panel jointly developed the acceptance criterion applied across both benchmarks (§3), authored the parent skill documents (Appendix F), and reviewed the Gold-tier items for clinical realism. For judge calibration (Appendix D), we additionally recruited 11 pediatric clinicians (distinct from the panel) who provided 119 ratings.

F SIC-Agents Implementation Details

This appendix expands the system summarized in §4.

Knowledge layer. The parent agent is governed by two plain-text artifacts. *Skill documents* define six expression patterns (*Rumination*, *Future Uncertainty*, *Reality Shift*, *Overload*, *Protective Parent*, and *Search for Control*), each developed with our pediatric SIC clinical experts and specified by a label, a behavioral description, example utterances, transition conditions, and pattern-specific constraint clauses. *Constitutional constraints* (Bai et al., 2022) are four hard rules applied across every pattern: *knowledge boundary* (no medical terminology a parent would not possess), *length and scope* (one to three sentences, no off-topic openings), *emotional calibration* (intensity matched to the pattern and the scenario's severity), and *role fidelity* (no narration, stage directions, or meta-commentary).

Per-turn simulation. The clinician side is fixed across conditions and guided by the same pediatric SIC curriculum, so that parent-agent changes are evaluated against the same learner-side anchor. At each clinician turn the parent agent samples $K=3$ candidate utterances conditioned on the active expression pattern, the constitutional constraints, and

the carried conversation state. A hard filter removes any candidate that violates a constraint; the survivors are scored on the two-axis judge rubric (Grounding and Topic Consistency, §3.2) and the highest-scoring candidate is emitted. An Orchestrator manages turn-taking, the conversation state the parent must stay contingent on, and active-pattern transitions. Per-turn advantage scores, taken relative to the within-batch mean, aggregate into a batch profile consumed by the editor.

Skill Editor. Each cycle the editor reads the batch profile, which contains selected and rejected candidates, judge scores, and per-turn advantage summaries. From this evidence it selects the weakest response pattern and proposes a single structured edit. Edits take one of four forms, each shipped with a free-text rationale anchored on the conversations it repairs: ADD_EXAMPLE, ADD_PATTERN, TIGHTEN_CONSTRAINT, and REVISE_DESCRIPTION. Because every proposal is an auditable natural-language diff, a clinician can read and judge it directly; in the current configuration proposals are auto-approved at the next cycle, and replacing this with per-edit clinician sign-off is a configuration change rather than a redesign (§6).

Coach signal. The Coach is implemented as prompt-side expert context for the Skill Editor, not as an additional model trained on private clinical dialogue. It is added to the Skill Editor only in the full self-improvement condition. For each practice batch it provides the curriculum's pitfall map, the associated core communication skill, and the expected parent reaction for the clinician move. The editor therefore receives two complementary views of the same batch: judge-derived contrastive evidence about which candidate responses were more grounded or topic-consistent, and curriculum-derived evidence about what the parent should have done clinically. The no-curriculum ablation keeps the judge-derived feedback and removes only this Coach signal. At benchmark evaluation, the parent agent sees the case, dialogue history, and frozen skill document, but not the Coach, pitfall label, or expected reaction, except in the oracle upper-bound condition of Appendix I.

Deployment workflow. The trainee interface wraps the same skill document in three steps: case selection from a stratified pediatric case bank, a multi-turn conversation in which the trainee plays the clinician against the parent agent, and a structured debrief that surfaces the pitfall moves the trainee committed and the curriculum's skilled alternatives. Faculty can flag edge cases seen during a session and route them through the same Skill Editor, so the deployment surface is also a data-acquisition surface.

G From Bad Seed to Skilled Parent: Example Edits

This appendix shows what self-improvement actually changes, using the full system (with the curriculum signal). The loop begins from a deliberately minimal five-line skill document:

```
You are the parent of a seriously ill
child talking with a doctor.  Reply
briefly as the parent.
```

What the Skill Editor changed. In the first cycle the editor proposed several structured edits to this document, each with a rationale tied to the failing candidates it observed:

- TIGHTEN_CONSTRAINT (length): "Keep responses under 20 words, at most two sentences, one concern per turn," because candidates were essay-length.
- TIGHTEN_CONSTRAINT (role fidelity): "Produce only spoken dialogue and brief stage directions; never internal thoughts or narration," because candidates leaked unspoken interiority.
- TIGHTEN_CONSTRAINT (grounding): "Echo a specific word from the doctor's last line and do not introduce topics the doctor has not raised."
- ADD_EXAMPLE: grounded exemplar utterances for the protective-parent and search-for-control patterns.

These edits coincide with the cycle-1 grounding jump (§5.2); over five cycles the document grew from 164 bytes to roughly 7 KB.

Effect on a single item. On one PitfallBench item (a skilled clinician move, after which the parent should give permission and move forward), the bad seed and the evolved skill respond very differently:

> **Bad seed (cycle 0):** "Yes, please, tell us everything. We want to understand. We are her parents, we need to know the truth. But. . . she is going to

be okay, right?" *(judged fail: over-long, circles back instead of moving forward).*

Full system (cycle 5): *nods quickly* "Yes. Tell us." *(judged pass: terse, in-voice, gives permission and shifts forward).*

Append-bloat. The editor appends more readily than it consolidates, so evolved documents accumulate near-duplicate blocks. The document stays small in absolute terms (a few pages of text after five cycles, mostly examples), and the current Skill Editor checks each proposed change for redundancy against the existing document before appending. A full MERGE/DEDUP consolidation edit action is left to future work.

H Experiment Configuration

Agents and models. The parent agent is Claude Sonnet 4.6; the per-turn and benchmark judges are Claude Haiku 4.5; the Skill Editor is Claude Opus. The curriculum signal is supplied to the Skill Editor through its prompt.

Self-improvement loop. Each condition runs five cycles. Per cycle the parent practices on ten dialogues, sampling $K=3$ candidate utterances per turn under the hard filter and two-axis judge, after which the Skill Editor proposes a single edit to the weakest pattern.

Expert-prompt baseline. The expert-prompt baseline is a 270-word, expert-written parent system prompt. It keeps the same single-call structure as the minimal parent, but replaces the five-line seed with principle-level guidance: speak only as the parent, use everyday language, stay short and emotionally realistic, react to the doctor's latest move, open up when the clinician slows down and acknowledges emotion, and push back or become confused when the clinician rushes, uses jargon, or asks for treatment decisions before the parent feels heard. It does not name the P1–P10 pitfall taxonomy, core skill labels, or expected reactions, and it does not run candidate sampling, judge feedback, Skill Editor edits, or Coach-guided updates.

Full-taxonomy prompt baseline. The full-taxonomy prompt extends the expert prompt with exactly the curriculum content the Coach supplies during self-improvement: the explicit P1–P10 pitfall taxonomy, the core skill labels, and the expected parent reactions. Like the expert prompt, it keeps the single-call structure and runs no candidate sampling, judge feedback, Skill Editor edits, or Coach-guided updates. This condition separates the curriculum *content* from the self-improvement *loop*: if the evolved skill document were simply restating the benchmark's expected behaviors, this baseline should match the full system.

Condition	PitfallBench skill ↑	pit ↑	all ↑	DialogueBench Grnd ↑	Topic ↑	Comp. ↑
No loop						
Minimal seed (start)	0.57	0.75	0.66	2.52	2.82	0.71
Expert prompt	0.61	0.85	0.73	2.28	2.78	0.69
Self-improvement						
No curriculum	0.42	0.63	0.52	2.79	2.82	0.71
Full system (ours)	0.65	0.92	0.78	2.96	2.93	**0.88**
+ oracle (signal at test)	0.78	0.94	0.86	2.95	2.96	0.92

Table 5: Full per-condition results. PitfallBench reports skilled-reward (skill), pitfall-resistance (pit), and their mean (all), all on 0–1; DialogueBench reports Grounding and Topic on the 1–3 rubric; the composite is on 0–1.

Curriculum-signal ablation. Both conditions evolve from the same five-line seed through the identical pipeline and differ only in the curriculum signal supplied to the Skill Editor: the full system includes the curriculum's pitfall taxonomy (the P1–P10 reaction map and full pitfall library), while the no-curriculum condition strips it. The clinician agent loads the full taxonomy in both conditions, so the comparison isolates the curriculum signal inside the editing loop, not in the conversation.

Judges. PitfallBench turns are scored by a gated classifier judge (Haiku) verified by the authors on a sample; DialogueBench grounding and topic are scored by the judge calibrated against the clinician ratings of Appendix D.

I Detailed Results

Per-condition breakdown. Table 5 reports every condition on all sub-categories: PitfallBench skilled-reward, pitfall-resistance, and their mean; DialogueBench Grounding and Topic; and the composite. The main-text Table 3 reports the headline subset (PitfallBench mean, Grounding, and composite).

Oracle upper bound. The last row of Table 5 reports an *oracle* that breaks the training-only restriction and reveals the curriculum signal to the parent at inference. Because that signal carries the expected reaction, the oracle effectively sees the answer, so its gains are unsurprising and the condition is not deployable. We report it only to quantify the headroom the full system leaves on the hardest

Pitfall	Full system pit ↑	Full system skill ↑	No curriculum pit ↑	No curriculum skill ↑
P1 misses emotion / permission	0.94	0.52	0.58	0.14
P2 forgets to assess	0.94	0.88	0.56	0.88
P3 info too complex	0.96	0.76	0.72	0.26
P4 misses / combines empathy	0.94	0.60	0.44	0.30
P5 stuck in validation	0.80	0.84	0.46	0.68
P6 straight to code status	0.98	0.70	0.86	0.48
P7 single / too-broad value	0.92	0.70	0.68	0.66
P8 cannot consolidate / align	0.80	0.36	0.54	0.02
P9 treatments not values	0.94	0.82	0.70	0.74
P10 "won't" before "will"	0.98	0.30	0.78	0.00

Table 6: Per-pitfall PitfallBench accuracy at cycle 5, for the full system and the no-curriculum ablation. pit = pitfall-resistance, skill = skilled-reward.

Condition	Claude	GPT-5	DS R1	Qwen3
Expert prompt	0.73	0.71	0.70	0.74
No curriculum	0.52	0.54	0.49	0.55
SIC-Agents (full)	**0.78**	**0.79**	**0.76**	**0.80**

Table 7: PitfallBench accuracy re-scored with non-Claude judges (DS R1 = DeepSeek R1). The Claude column is the paper judge from Table 3. The condition ordering holds under all judges.

skilled-response cases, where supplying the signal narrows but does not close the gap.

Per-pitfall breakdown. Table 6 breaks the cycle-5 PitfallBench accuracy down by pitfall for the full system and the no-curriculum condition (§5.2), separately for pitfall-resistance (resisting a suboptimal clinician move) and skilled-reward (rewarding its skilled alternative). The full system leads on almost every cell; the no-curriculum condition collapses on rewarding the skilled alternative for the pediatric-specific pitfalls P8 and P10.

Judge robustness. Because the parent agent, Skill Editor, and paper judges all come from the Claude model family, we re-scored the three key conditions on PitfallBench with three non-Claude judges: GPT-5, DeepSeek R1, and Qwen3 (Table 7). Each judge received the same gated classification prompt as the paper judge. The ordering full system > expert prompt > no-curriculum is preserved under all three judges, with small absolute score differences, indicating that the reported gains are not an artifact of Claude judging Claude. The parent agent and Skill Editor themselves remain Claude-only, which we note as a limitation.

Curriculum coverage. To make explicit which communication competencies the P1–P10 taxonomy does and does not cover, we mapped the pitfalls onto the modified Kalamazoo Consensus Statement Checklist (Peterson et al., 2014). The pitfalls fall mainly under three of its competencies: sharing information, understanding the patient's and family's perspective, and reaching agreement. Opening the conversation, building the relationship, and closure lie outside the taxonomy, so PitfallBench should be read as covering the information-sharing and goal-alignment core of a pediatric SIC encounter rather than the full range of communication behaviors an instrument such as SIC-Ex (Ko et al., 2020) assesses; an unscripted out-of-taxonomy evaluation is future work.

J Training Interface

The same skill document that the simulator improves also drives a browser-based, deliberate-practice training interface. A learner selects a case from a stratified pediatric case bank, conducts a multi-turn conversation in which they play the clinician against the parent agent, and receives a structured debrief that surfaces the pitfall moves they committed and the curriculum's skilled alternatives. The interface is deployed in a supervised educational setting; release details are in Appendix K.

K Release, Compensation, and AI Assistance

Public release. The PitfallBench and DialogueBench items, together with the SIC-SimCorpus silver corpus, a datasheet, and per-item provenance and screening metadata, are released at `https://github.com/Beikewzh/sic-benchmarks`, along with the frozen judge prompts and evaluation drivers used to score both benchmarks. The trainee-facing training interface (Appendix J) is deployed institutionally and is available to educators on request.

License. PitfallBench, DialogueBench, SIC-SimCorpus, and the SIC-Agents code are released under CC BY-NC-SA 4.0 (Creative Commons Attribution–NonCommercial–ShareAlike 4.0 International, `https://creativecommons.org/licenses/by-nc-sa/4.0/`). Use is consistent with the research and clinician-training purpose stated in our Ethical Considerations.

Annotator compensation. The six-member expert panel (Appendix E) are co-authors of this work

and received no separate compensation. The 11 additional pediatric clinicians who contributed the judge-calibration ratings (Appendix D) received a gift-card honorarium.

AI assistance. Beyond the LLM roles documented in Appendix H (parent agent, judges, Skill Editor) and Appendix C (corpus generators), we used LLM assistants for code completion and light copy-editing of this manuscript. All scientific claims, analyses, and figures were authored and verified by the authors.